

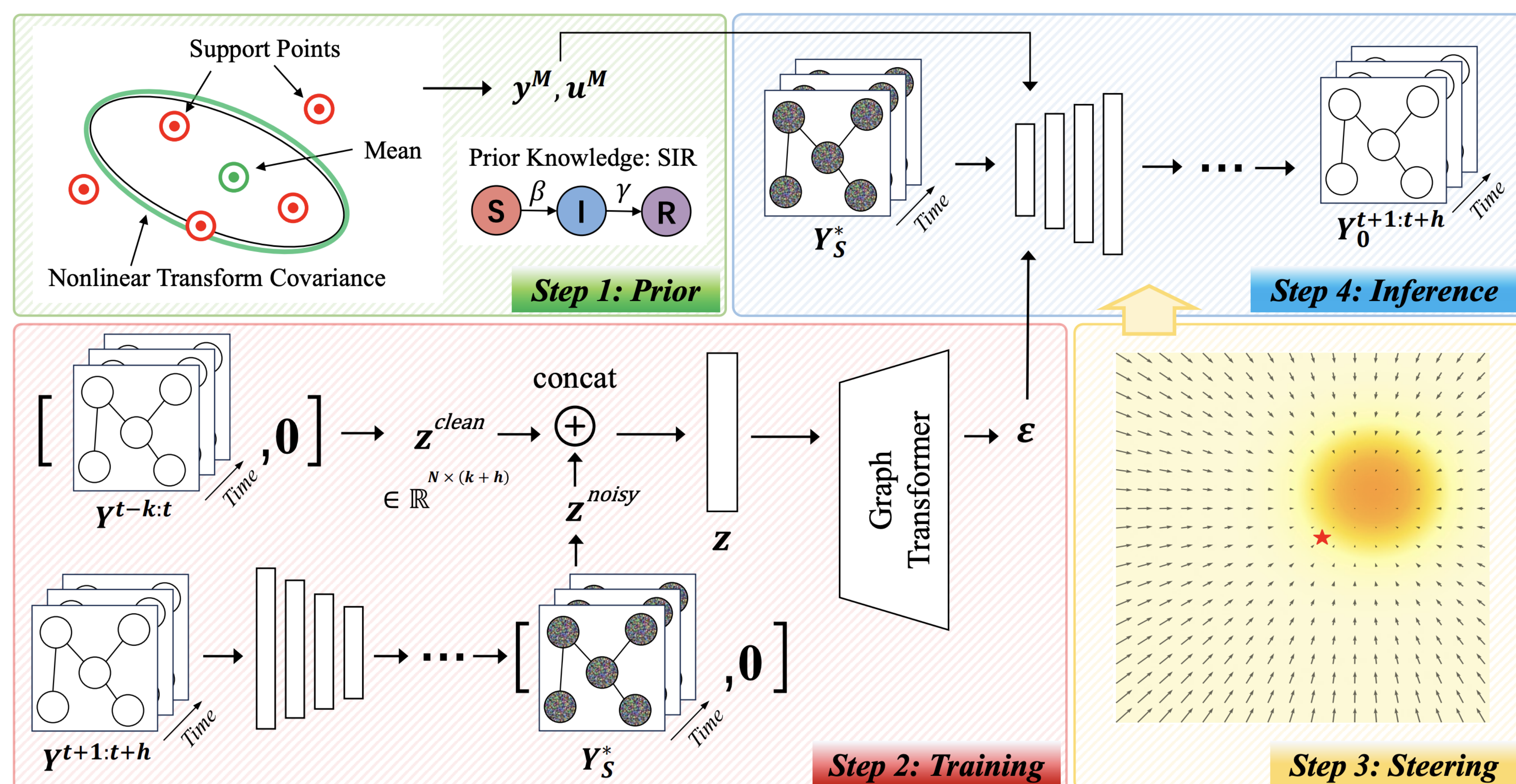


Motivation

- Reliable epidemic forecasting is critical for public health decisions, but real-world outbreaks often involve sparse data and non-stationary dynamics.
- Mechanistic models are interpretable but can be mis-specified, while data-driven models are flexible but fragile under distribution shifts.
- Existing hybrid methods often fuse mechanistic priors as auxiliary signals, without explaining when the model should trust mechanistic dynamics versus data-driven patterns.

How can we turn imperfect epidemic dynamics into uncertainty-calibrated guidance for generative forecasting?

Proposed Framework: EpiDiff



① **Mechanistic Prior Estimation**: Fit SIR dynamics to historical trajectories and estimate parameter uncertainty.

$$p(\theta_i | y_i^{t-\kappa:t-1}) \approx N(\hat{\theta}_i, H_i^{-1})$$

② **Uncertainty Propagation**: Generate mechanistic forecasts and uncertainty by propagating support points through epidemic dynamics.

$$y_{i,t+1}^M = \sum_m w^m \phi(y_i^t, \theta_i^m), \quad u_{i,t+1}^M = \sum_m w^m (y_{i,t+1}^m - y_{i,t+1}^M)^2$$

③ **Diffusion Forecasting Backbone**: Train a spatiotemporal graph diffusion model to generate future epidemic trajectories from noisy futures, conditioned on historical observations and graph structure.

④ **Uncertainty-Aware Steering**: During inference, treat the diffusion model as a data-driven prior and the mechanistic estimate as an uncertainty-weighted physical likelihood.

$$p_{\text{post}}(y_0) \propto p_{\psi}(y_0 | Y_{t-\kappa:t}, G) \exp \left[-\frac{\tau}{2} (y_0 - y^M)^T \Sigma_M^{-1} (y_0 - y^M) \right]$$

Core idea: EpiDiff turns imperfect epidemic dynamics into uncertainty-calibrated guidance for diffusion forecasting.

Contributions

- Unified Hybrid Framework**. We propose EpiDiff, a diffusion-based spatiotemporal forecasting framework that integrates mechanistic epidemic dynamics with data-driven generation.
- Uncertainty-Aware Guidance**. We quantify the uncertainty of mechanistic estimates and use it to dynamically steer the diffusion process, enabling adaptive reliance on mechanistic versus data-driven signals.
- Empirical Effectiveness**. On real-world COVID-19 and influenza datasets, EpiDiff consistently improves forecasting accuracy and robustness while providing interpretable uncertainty-aware predictions.

Experiments

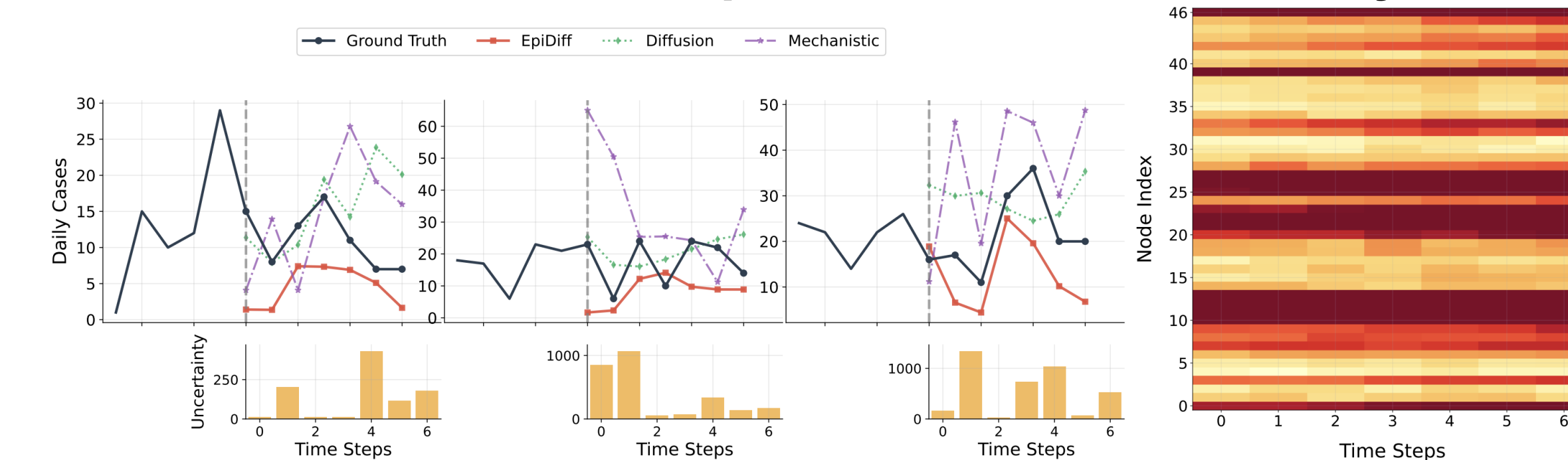
Performance comparison

Methods	covid-US				covid-JP				influenza-US			
	$h=7$	$h=14$	$h=7$	$h=14$	$h=7$	$h=14$	$h=7$	$h=14$	$h=7$	$h=14$	$h=7$	$h=14$
PatchSIR	1791±0	891±0	2077±0	886±0	313±0	98±0	332±0	106±0	323±0	82±0	454±0	138±0
AR	1791±16	835±16	1943±23	865±7	457±57	159±11	476±11	158±4	4579±2429	2040±1814	2328±236	1172±84
ARMA	1832±29	843±2	1906±18	852±9	460±32	156±8	496±6	166±3	4423±1971	2289±1271	2314±150	1174±68
LSTM	1797±59	867±48	1956±49	962±52	408±10	135±3	398±19	135±7	353±29	119±21	354±42	133±7
GRU	1654±81	771±52	1949±38	974±30	364±12	120±3	383±8	130±3	377±41	142±13	498±303	192±75
STGCN	2131±12	877±23	2446±51	1107±12	315±83	96±23	326±24	103±9	414±93	160±42	588±128	248±56
ColaGNN	2740±758	1175±339	3055±356	1403±181	213±20	63±4	305±47	88±11	250±17	85±9	347±228	129±27
STGODE	1763±96	719±34	2716±107	1010±40	269±10	79±2	309±23	96±7	643±194	261±94	1415±460	570±200
DiffSTG	1236±28	443±11	1284±20	444±14	246±28	81±9	423±35	147±13	344±29	151±69	520±44	254±54
STAN	1364±196	1150±174	1265±98	1046±90	222±87	127±40	224±88	128±41	4599±42	4306±32	3750±79	3270±79
EARTH	2145±324	603±103	2664±272	794±79	303±73	100±13	394±102	134±21	539±172	190±47	2027±1160	629±302
EpiColaGNN	1825±84	745±58	2653±22	1235±9	282±33	83±11	327±19	99±6	1311±67	693±36	1514±64	807±36
EpiDiff	1096±5	374±1	1232±5	422±16	188±7	62±2	247±2	86±1	304±1	82±0.4	423±1	125±0.1

Ablation

Variants	covid-US				covid-JP				influenza-US			
	$h=7$	$h=14$	$h=7$	$h=14$	$h=7$	$h=14$	$h=7$	$h=14$	$h=7$	$h=14$	$h=7$	$h=14$
w/o Diffusion	1791	891	2077	886	313	98	332	106	323	82	454	138
w/o Steering	1080±12	348±1	1199±15	386±11	370±12	94±5	417±4	121±1	438±19	224±7	1787±87	928±59
w/o Uncertainty	1099±15	376±1	1253±15	438±18	316±15	97±1	365±1	122±1	328±1	147±1	446±10.4	155±10.3
EpiDiff	1096±5	374±1	1232±5	422±16	188±7	62±2	247±2	86±1	304±1	82±0.4	423±1	125±0.1

Mechanistic Interpretation and Uncertainty



Observations

- EpiDiff consistently outperforms strong mechanistic, time-series, and spatiotemporal graph baselines across COVID-19 and influenza datasets.
- Uncertainty-aware steering is especially effective under non-stationary dynamics, where the diffusion backbone alone struggles to track shifted epidemic trends.
- Mechanistic uncertainty explains model reliance: confident priors steer forecasts toward epidemic dynamics, while high uncertainty lets diffusion dominate.

Acknowledgments

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